Notes on Computational Fluid Dynamics: General Principles

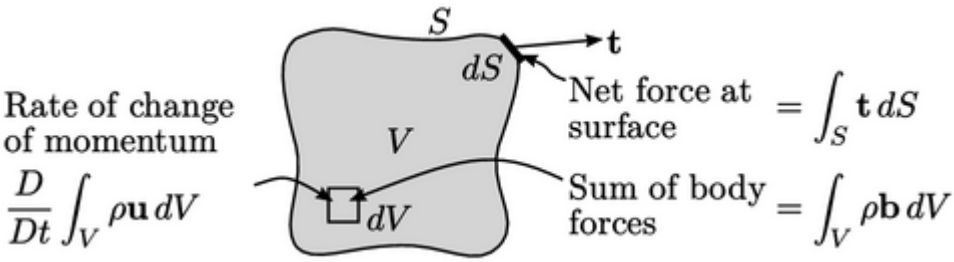
2.7 Conservation of momentum

The law of conservation of momentum can be written[4]

$$\rho \frac{D\mathbf{u}}{Dt} = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{b} \tag{2.19}$$

where $\mathbf{b}$ represents any body force per unit mass. Body forces represent any force which does not act at a bounding surface, including those that act at a distance, such as gravitational force.

The equation is derived by considering the time rate of change of the momentum of a mass of particles. We consider a *volume of material of fixed mass* moving through space and therefore present rate of change by the material derivative D/Dt.



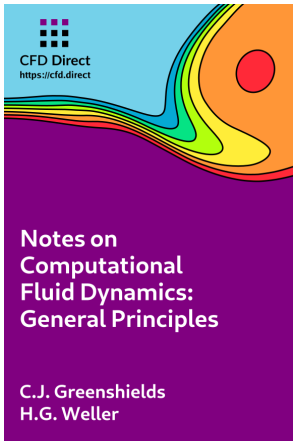
Applying Gauss's theorem and Eq. (2.16), the surface force is

$$\int_S \mathbf{t}\, dS = \int_S \mathbf{n} \cdot \boldsymbol{\sigma}\, dS = \int_S (d\mathbf{S} \cdot \boldsymbol{\sigma}) = \int_V \nabla \cdot \boldsymbol{\sigma}\, dV\,. \tag{2.20}$$

Equating the rate of change of momentum to the forces and, noting mass is fixed so $\rho\, dV$ is constant in time, gives

$$\int_V \left[\rho \frac{D\mathbf{u}}{Dt} - \nabla \cdot \boldsymbol{\sigma} - \rho \mathbf{b} \right] dV = 0\,. \tag{2.21}$$

The integrand must equal 0, resulting in Eq. (2.19).



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Divergence was described in section 2.4 as the flux across a surface per unit volume as $\Delta V \to 0$. The divergence of stress similarly represents the stress flux across the surface, *i.e.* force, per unit volume as $\Delta V \to 0$, given by

$$\nabla \bullet \boldsymbol{\sigma} = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \int_S (d\mathbf{S} \bullet \boldsymbol{\sigma}). \tag{2.22}$$

Eq. (2.19) specifically relates to *linear* momentum. Instead, conservation of *angular* momentum[5], in the absence of any "couple stresses" that generate a moment field, is given by

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\mathrm{T}} \quad (\boldsymbol{\sigma} \text{ is symmetric}). \tag{2.23}$$

The derivation of Eq. (2.23) is fairly complex so is omitted here.

Tensor symmetry

Symmetry in a tensor $\mathbf{T}$ refers to components being symmetric about the diagonal, *i.e.* $T_{12} = T_{21}$, $T_{13} = T_{31}$ and $T_{23} = T_{32}$. The *transpose* of a tensor, denoted by the 'T' superscript, switches components across the diagonal such that:

$$\mathbf{T}^{\mathrm{T}} = \begin{pmatrix} T_{xx} & T_{yx} & T_{zx} \\ T_{xy} & T_{yy} & T_{zy} \\ T_{xz} & T_{yz} & T_{zz} \end{pmatrix} \tag{2.24}$$

$\mathbf{T}$ is therefore *symmetric* if $\mathbf{T} = \mathbf{T}^{\mathrm{T}}$. A *skew* (anti-symmetric) tensor has $\mathbf{T} = -\mathbf{T}^{\mathrm{T}}$. A tensor can be decomposed into symmetric and skew parts by:

$$\mathbf{T} = \underbrace{\frac{1}{2}(\mathbf{T} + \mathbf{T}^{\mathrm{T}})}_{symmetric} + \underbrace{\frac{1}{2}(\mathbf{T} - \mathbf{T}^{\mathrm{T}})}_{skew} = \mathrm{sym}\,\mathbf{T} + \mathrm{skw}\,\mathbf{T} \tag{2.25}$$

[4] Cauchy's first law of motion (1827)

[5] Cauchy's second law of motion (1827)

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